



IBM Bob's Analysis

✓ **Success!** Assignment analyzed successfully! IBM Bob has generated your personalized study plan.

Your assignment has been decoded and analyzed!



Task Checklist

Estimated total time: 4 hours 25 minutes

Create Student model class with id, name, email, age fields and validation annotations

🕒 20 minutes | 🔥 Priority: HIGH

Implement StudentController with @RestController and @RequestMapping

🕒 15 minutes | 🔥 Priority: HIGH

Create GET /api/students endpoint to retrieve all students

🕒 15 minutes | 🔥 Priority: HIGH

Create GET /api/students/{id} endpoint for single student retrieval

🕒 10 minutes | 🔥 Priority: HIGH

Create POST /api/students endpoint with @Valid annotation for creating students

🕒 20 minutes | 🔥 Priority: HIGH

Create PUT /api/students/{id} endpoint for updating student information

🕒 20 minutes | 🔥 Priority: MEDIUM

Create DELETE /api/students/{id} endpoint for removing students

🕒 15 minutes | 🔥 Priority: MEDIUM

Add @Email validation for email field in Student model

🕒 5 minutes | 🔥 Priority: HIGH

Add @Min and @Max validation for age field (18-100)

🕒 5 minutes | 🔥 Priority: HIGH

Create StudentService layer for business logic

🕒 25 minutes | 🔥 Priority: MEDIUM

Implement exception handling with @ControllerAdvice

🕒 20 minutes | 🔥 Priority: MEDIUM

Write unit tests for StudentController using MockMvc

🕒 30 minutes | 🔥 Priority: HIGH

Write unit tests for StudentService with Mockito

🕒 25 minutes | 🔥 Priority: HIGH

Add integration tests for REST endpoints

🕒 30 minutes | 🔥 Priority: MEDIUM

Configure H2 in-memory database for testing

🕒 10 minutes | 🔥 Priority: LOW



Generated Code Stubs

📋 Copy Code

Bob has generated starter code with TODOs for you to complete:

```
// Sample generated code
package com.example.studentapi.model;

import jakarta.validation.constraints.*;

/**
 * Student entity - Generated by IBM Bob
 */
public class Student {
    private Long id;

    @NotBlank(message = "Name is required")
    private String name;

    @Email(message = "Email should be valid")
```

```
private String email;
```

```
@Min(value = 18)
```

```
@Max(value = 100)
```



Generated Test Cases

[Copy Tests](#)

Bob has created comprehensive JUnit 5 tests:

```
import org.junit.jupiter.api.*;
import static org.junit.jupiter.api.Assertions.*;

/**
 * Test cases - Generated by IBM Bob
 */
class StudentTest {

    @Test
    @DisplayName("Should validate correct email format")
    void testValidEmail() {
        Student student = new Student();
        student.setEmail("test@example.com");
        assertTrue(student.getEmail().contains("@"));
    }

    @Test
    @DisplayName("Should reject age below 18")
    void testAgeValidation() {
        Student student = new Student();
```



IBM Bob's Study Schedule

[Copy Schedule](#)

 Study Schedule - Assignment Completion Plan

****Assignment:**** Student Management System REST API

****Due Date:**** 2026-05-17

****Total Estimated Time:**** 3 hours 30 minutes

****Recommended Daily Study:** 2 hours**

Day 1 (Today) - Foundation & Core Models

****Focus:** Set up project structure and create data models**

- ☒ Create Student model class (20 min)
- ☒ Add validation annotations (@Email, @Min, @Max) (10 min)
- ☒ Implement StudentController skeleton (15 min)
- ☒ Create StudentService layer (25 min)

****Total:** 1 hour 10 minutes**

Day 2 - CRUD Operations

****Focus:** Implement all REST endpoints**

- ☒ Implement GET /api/students endpoint (15 min)
- ☒ Implement GET /api/students/{id} endpoint (10 min)
- ☒ Implement POST /api/students endpoint (20 min)
- ☒ Implement PUT /api/students/{id} endpoint (20 min)
- ☒ Implement DELETE /api/students/{id} endpoint (15 min)
- ☒ Add exception handling with @ControllerAdvice (20 min)

****Total:** 1 hour 40 minutes**

Day 3 - Testing & Polish

****Focus:** Achieve 80% test coverage**

- ☒ Write unit tests for StudentController (30 min)
- ☒ Write unit tests for StudentService (25 min)
- ☒ Add integration tests (30 min)
- ☒ Configure H2 database (10 min)
- ☒ Final testing and bug fixes (20 min)

****Total:** 1 hour 55 minutes**

💡 Pro Tips from IBM Bob:

1. ****Start with models**** - Get your data structure right first
2. ****Test as you go**** - Don't wait until the end to write tests
3. ****Use Spring Boot DevTools**** - Auto-restart saves time
4. ****Commit frequently**** - Use Git to track your progress
5. ****Take breaks**** - 25-minute focused sessions work best

🎯 Success Metrics:

- [] All CRUD endpoints working
- [] Email validation functional
- [] Age validation (18-100) working
- [] 80%+ test coverage achieved
- [] No compilation errors
- [] API documented (Swagger/OpenAPI)

****Estimated completion:** 3 days before due date ✨**



Generated Files

Bob has generated code stubs and tests at: `.\temp\1778921111661\generated-code`
Test files at: `.\temp\1778921111661\generated-tests`

 [View Bob Interaction Evidence](#) [Analyze Another Assignment](#) [Print Results](#)

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